Quantum Mechanics 2

Worked Solutions

BPhO Computational Physics Challenge 2026

Introduction

This document gives worked solutions to the *Quantum Mechanics 2* problem sheet. The calculations use the constants supplied on the sheet and have been checked against the accompanying BPhO solution document. The aim is to show the physical reasoning, mathematical derivations and numerical substitutions rather than simply state the final answers.

1 Question 1 — The Caesium-133 clock transition

The definition of the SI second is based on radiation from Caesium-133 with frequency

$$f = 9\,192\,631\,770 \text{ Hz}.$$

1.1 (i) Wavelength

For electromagnetic radiation,

$$c = f\lambda,$$

so

$$\lambda = \frac{c}{f}.$$

Using

$$c = 2.998 \times 10^8 \text{ m s}^{-1},$$

we obtain

$$\lambda = \frac{2.998 \times 10^8}{9.192631770 \times 10^9} \approx 3.26 \times 10^{-2} \text{ m}.$$

Thus

$$\boxed{\lambda \approx 3.26 \text{ cm}}.$$

A wavelength of a few centimetres corresponds to the

$$\boxed{\text{microwave}}$$

region of the electromagnetic spectrum.

1.2 (ii) Energy change

A photon has energy

$$E = hf.$$

The energy in electron-volts is

$$E_{\text{eV}} = \frac{hf}{e}.$$

Therefore

$$E_{\mathrm{eV}} = \frac{(6.63 \times 10^{-34})(9.192631770 \times 10^{9})}{1.60 \times 10^{-19}} \approx 3.81 \times 10^{-5}\ \mathrm{eV}.$$

Hence

$$\boxed{\Delta E \approx 3.81 \times 10^{-5}\ \mathrm{eV}}.$$

This is an *electronic* rather than a nuclear energy scale. The relevant transition is between very closely spaced atomic states of Caesium; nuclear transitions typically involve energies of order keV–MeV, vastly larger than this microwave transition.

2 Question 2 — Photoelectric effect

The photoelectric equation is

$$K_{\max} = hf - \phi,$$

where ϕ is the work function.

2.1 (i) Work function

No electrons are emitted for wavelengths greater than 300 nm. Thus the threshold wavelength is

$$\lambda_0 = 300\,\mathrm{nm}.$$

At threshold the emitted electron has zero kinetic energy:

$$hf_0 = \phi.$$

Since

$$f_0 = \frac{c}{\lambda_0},$$

we obtain

$$\phi = \frac{hc}{\lambda_0}.$$

Thus

$$\phi = \frac{(6.63 \times 10^{-34})(2.998 \times 10^{8})}{300 \times 10^{-9}} \approx 6.63 \times 10^{-19}\ \mathrm{J}.$$

Converting to electron-volts,

$$\phi = \frac{6.63 \times 10^{-19}}{1.60 \times 10^{-19}} \approx 4.14\ \mathrm{eV}.$$

Therefore

$$\boxed{\phi \approx 4.14\ \mathrm{eV}}.$$

2.2 (ii) Photoelectron current

The incident laser has power

$$P = 100\ \mathrm{W}$$

and wavelength

$$\lambda = 200\,\mathrm{nm}.$$

The energy of each incident photon is

$$E_\gamma = \frac{hc}{\lambda}.$$

Hence

$$E_\gamma = \frac{(6.63 \times 10^{-34})(2.998 \times 10^{8})}{200 \times 10^{-9}} \approx 9.94 \times 10^{-19} \text{ J}$$

or

$$E_\gamma \approx 6.21 \text{ eV}.$$

The maximum kinetic energy of an emitted electron is therefore

$$K_{\text{max}} = E_\gamma - \phi \approx 6.21 - 4.14 = 2.07 \text{ eV}.$$

In joules,

$$K_{\text{max}} \approx 3.31 \times 10^{-19} \text{ J}.$$

The problem states that the photoelectron current constitutes 10% of the incident power, so the power carried by the emitted electrons is

$$P_e = 0.10(100) = 10 \text{ W}.$$

Thus the electron emission rate is

$$\dot{N}_e = \frac{P_e}{K_{\text{max}}} \approx \frac{10}{3.31 \times 10^{-19}} \approx 3.02 \times 10^{19} \text{ s}^{-1}.$$

Assuming each electron contributes one elementary charge,

$$I = e\dot{N}_e.$$

Therefore

$$I = (1.60 \times 10^{-19})(3.02 \times 10^{19}) \approx 4.83 \text{ A}.$$

Hence

$$\boxed{I \approx 4.8 \text{ A}}.$$

3 Question 3 — de Broglie wavelength of a water molecule

The problem gives the molecular kinetic energy as

$$E = \frac{3}{2}k_B T.$$

For water,

$$\text{H}_2\text{O} = {}^{1}_{1}\text{H}_2 + {}^{16}_{8}\text{O}.$$

Ignoring electron masses and binding-energy mass defects, the mass is

$$m = 2m_p + 8m_p + 8m_n.$$

Thus

$$m = 10m_p + 8m_n.$$

Using

$$m_p = 1.6726 \times 10^{-27} \text{ kg}, \qquad m_n = 1.6749 \times 10^{-27} \text{ kg},$$

gives

$$m \approx 3.0125 \times 10^{-26} \text{ kg}.$$

At

$$T = 300 \text{ K},$$

the kinetic energy is

$$E = \frac{3}{2}(1.38 \times 10^{-23})(300) \approx 6.21 \times 10^{-21} \text{ J}.$$

Using

$$E = \frac{p^2}{2m},$$

the momentum is

$$p = \sqrt{2mE}.$$

The de Broglie relation gives

$$\lambda = \frac{h}{p},$$

so

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m(3k_BT/2)}}.$$

Therefore

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2(3.0125 \times 10^{-26})(6.21 \times 10^{-21})}} \approx 3.43 \times 10^{-11} \text{ m}.$$

Thus

$$\boxed{\lambda \approx 3.43 \times 10^{-11} \text{ m}}$$

or

$$\boxed{\lambda \approx 0.0343 \text{ nm}}.$$

This is much smaller than the size of an individual water molecule, illustrating why the wave nature of matter is difficult to observe for macroscopic objects.

4 Question 4 — Relativistic electrons

The electron is accelerated until

$$v = \frac{c}{2}.$$

4.1 (i) Accelerating voltage

Classical calculation

Classically,

$$K = \frac{1}{2}m_e v^2.$$

Since an electron accelerated through a potential difference V gains energy eV,

$$eV = \frac{1}{2}m_e v^2.$$

Setting $v = c/2$,

$$eV = \frac{1}{2}m_e \frac{c^2}{4} = \frac{m_e c^2}{8}.$$

Therefore

$$V = \frac{m_e c^2}{8e}.$$

Substitution gives

$$V \approx 6.40 \times 10^4 \text{ V}.$$

Hence

$$\boxed{V_{\text{classical}} \approx 64 \text{ kV}}.$$

Relativistic calculation

The relativistic kinetic energy is

$$K = (\gamma - 1)m_e c^2,$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

For $v = c/2$,

$$\gamma = \frac{1}{\sqrt{1 - \frac{1}{4}}} = \frac{2}{\sqrt{3}} \approx 1.155.$$

Therefore

$$eV = (\gamma - 1)m_e c^2,$$

so

$$V = \left(\frac{2}{\sqrt{3}} - 1\right)\frac{m_e c^2}{e}.$$

Numerically,

$$\boxed{V_{\text{relativistic}} \approx 79.2\ \text{kV}}.$$

The relativistic result is noticeably larger because the classical expression underestimates the kinetic energy at this speed.

4.2 (ii) de Broglie wavelength

Classical momentum

Classically,

$$p_{\text{cl}} = m_e v.$$

Thus

$$\lambda_{\text{cl}} = \frac{h}{m_e v}.$$

For $v = c/2$,

$$\lambda_{\text{cl}} = \frac{6.63 \times 10^{-34}}{(9.109 \times 10^{-31})(c/2)} \approx 4.86 \times 10^{-12}\ \text{m}.$$

Therefore

$$\boxed{\lambda_{\text{cl}} \approx 4.86 \times 10^{-12}\ \text{m}}.$$

Relativistic momentum

The relativistic momentum is

$$p = \gamma m_e v.$$

Hence

$$\lambda_{\text{rel}} = \frac{h}{\gamma m_e v}.$$

Using $\gamma = 2/\sqrt{3}$,

$$\lambda_{\text{rel}} \approx 4.20 \times 10^{-12}\ \text{m}.$$

Thus

$$\boxed{\lambda_{\text{rel}} \approx 4.20 \times 10^{-12}\ \text{m}}.$$

The relativistic wavelength is shorter because the relativistic momentum is larger than the classical value at the same velocity.

4.3 (iii) Atomic lattice spacing

The electron beam is diffracted through an angle

$$\theta = 2^\circ.$$

For first-order constructive interference,

$$d \sin\theta = n\lambda,$$

with

$$n = 1.$$

Thus

$$d = \frac{\lambda}{\sin 2^\circ}.$$

Using the more accurate relativistic wavelength,

$$d = \frac{4.20 \times 10^{-12}}{\sin 2^\circ} \approx 1.20 \times 10^{-10} \text{ m}.$$

Therefore

$$\boxed{d \approx 1.2 \times 10^{-10} \text{ m}}.$$

This is of the expected order of magnitude for an atomic lattice spacing.

5 Question 5 — Ionisation energies and the Bohr model

For a hydrogenic atom with nuclear charge $+Ze$ and one electron, the Bohr model gives

$$E_n = -\frac{13.6Z^2}{n^2} \text{ eV}.$$

Ionisation from the ground state therefore requires

$$E_{\text{ion}} = 13.6Z^2 \text{ eV}.$$

For a photon of energy E,

$$E = \frac{hc}{\lambda}.$$

Using $hc \approx 1242$ eV nm,

$$\lambda_{\max} = \frac{1242}{E_{\text{ion}}} \text{ nm}.$$

5.1 (i) Maximum ionising wavelength

(a) Hydrogen

For $Z = 1$,

$$E_{\text{ion}} = 13.6 \text{ eV}.$$

Therefore

$$\lambda_{\max} = \frac{1242}{13.6} \approx 91.3 \text{ nm}.$$

Thus

$$\boxed{\lambda_{\max}(\text{H}) \approx 91.3 \text{ nm}}.$$

(b) Helium

For the one-electron hydrogenic approximation with $Z = 2$,

$$E_{\text{ion}} = 13.6(2)^2 = 54.4 \text{ eV}.$$

Hence

$$\lambda_{\text{max}} = \frac{1242}{54.4} \approx 22.8 \text{ nm}.$$

Therefore

$$\boxed{\lambda_{\text{max}}(\text{He}) \approx 22.8 \text{ nm}}.$$

This is specifically the hydrogenic approximation; a real neutral helium atom has two electrons, so the assumptions of the model do not apply directly.

(c) Radon

For $Z = 86$,

$$E_{\text{ion}} = 13.6(86)^2 \approx 1.006 \times 10^5 \text{ eV}.$$

Thus

$$\lambda_{\text{max}} = \frac{1242}{1.006 \times 10^5} \approx 1.24 \times 10^{-2} \text{ nm}.$$

Therefore

$$\boxed{\lambda_{\text{max}}(\text{Rn}) \approx 0.0124 \text{ nm}}.$$

Again, this is the formal hydrogenic prediction. For multi-electron atoms the electron–electron repulsion and screening make the one-electron Bohr model increasingly inappropriate.

5.2 (ii)(a) Comparison with measured ionisation energies

The measured molar ionisation energies are

$$\begin{array}{lcl} \text{H} & : & 1312.0 \text{ kJ mol}^{-1}, \\ \text{He} & : & 2372.3 \text{ kJ mol}^{-1}, \\ \text{Rn} & : & 1037 \text{ kJ mol}^{-1}. \end{array}$$

The hydrogen value is close to the Bohr prediction because hydrogen is genuinely a one-electron atom. For helium and radon, however, the simple $13.6Z^2$ prediction becomes increasingly poor.

The fundamental issue is that the Bohr expression used above assumes a *hydrogenic, one-electron atom*: the electron is moving in the Coulomb field of a nucleus with charge $+Ze$, without accounting for the other electrons.

A measured ionisation energy for a neutral atom instead means removing one electron from the actual multi-electron atom. Electron–electron repulsion and shielding therefore have a major effect.

For example, the formal Bohr prediction for radon is

$$\sim 100.6 \text{ keV},$$

whereas its measured first ionisation energy is only about

$$10.8 \text{ eV}.$$

This enormous discrepancy demonstrates the failure of applying the hydrogenic formula directly to a many-electron atom.

5.3 (ii)(b) Photon wavelengths for the measured ionisation energies

The energy per atom corresponding to a molar ionisation energy E_{mol} is

$$E = \frac{E_{\text{mol}}}{N_A}.$$

Equivalently, converting directly to electron-volts,

$$E_{\text{eV}} = \frac{E_{\text{mol}} \times 10^3}{N_A e}.$$

The wavelength is then

$$\lambda = \frac{hc}{E}.$$

Hydrogen

$$E = \frac{1312.0 \times 10^3}{6.022140857 \times 10^{23}} \approx 2.18 \times 10^{-18} \text{ J},$$

giving

$$\boxed{\lambda_{\text{H}} \approx 91.2 \text{ nm}}.$$

Helium

$$E = \frac{2372.3 \times 10^3}{6.022140857 \times 10^{23}},$$

giving

$$\boxed{\lambda_{\text{He}} \approx 50.5 \text{ nm}}.$$

Radon

$$E = \frac{1037 \times 10^3}{6.022140857 \times 10^{23}},$$

giving

$$\boxed{\lambda_{\text{Rn}} \approx 115 \text{ nm}}.$$

The measured wavelengths therefore differ dramatically from the hydrogenic prediction for the heavy, many-electron atom.

6 Question 6 — FAST and photons from Andromeda

FAST has diameter

$$D = 500 \text{ m},$$

so its collecting area is

$$A = \pi \left(\frac{D}{2}\right)^2 = \pi(250)^2.$$

The distance to Andromeda is

$$r = 2 \times 10^{22} \text{ m},$$

and the power radiated at

$$f = 1420 \text{ MHz} = 1.420 \times 10^9 \text{ Hz}$$

is

$$P_{\text{A}} = 8 \times 10^{27} \text{ W}.$$

6.1 (i) Photons received by FAST

Assuming isotropic radiation, the power flux at FAST is

$$I_{\mathrm{A}} = \frac{P_{\mathrm{A}}}{4\pi r^2}.$$

The power intercepted by FAST is

$$P_{\mathrm{FAST}} = I_{\mathrm{A}} A.$$

Combining these,

$$P_{\mathrm{FAST}} = \frac{P_{\mathrm{A}}}{4\pi r^2}\pi\left(\frac{D}{2}\right)^2 = \frac{P_{\mathrm{A}}}{4}\left(\frac{D}{2r}\right)^2.$$

Numerically,

$$P_{\mathrm{FAST}} \approx 3.13 \times 10^{-13}\ \mathrm{W}.$$

Each photon has energy

$$E_\gamma = hf.$$

Thus

$$E_\gamma = (6.63 \times 10^{-34})(1.420 \times 10^9) \approx 9.41 \times 10^{-25}\ \mathrm{J}.$$

Therefore the number of photons received per second is

$$\dot{N}_{\mathrm{FAST}} = \frac{P_{\mathrm{FAST}}}{hf} \approx 3.3 \times 10^{11}\ \mathrm{s}^{-1}.$$

Hence

$$\boxed{\dot{N}_{\mathrm{FAST}} \approx 3.3 \times 10^{11}\ \mathrm{photons\,s^{-1}}}.$$

6.2 (ii) DSLR exposure

The DSLR sensor has dimensions

$$24\,\mathrm{mm} \times 36\,\mathrm{mm},$$

so its area is

$$A_{\mathrm{DSLR}} = (24 \times 10^{-3})(36 \times 10^{-3}) = 8.64 \times 10^{-4}\ \mathrm{m}^2.$$

With solar intensity

$$I_\odot = 100\ \mathrm{W\,m^{-2}},$$

the incident power is

$$P_{\mathrm{DSLR}} = I_\odot A_{\mathrm{DSLR}} = 0.0864\ \mathrm{W}.$$

For a shutter time of

$$t = \frac{1}{500}\ \mathrm{s},$$

the received energy is

$$E_{\mathrm{DSLR}} = P_{\mathrm{DSLR}} t = 1.728 \times 10^{-4}\ \mathrm{J}.$$

For sunlight of wavelength

$$\lambda = 500\ \mathrm{nm},$$

the photon energy is

$$E_\gamma = \frac{hc}{\lambda} \approx 3.98 \times 10^{-19}\ \mathrm{J}.$$

Thus the number of photons in the exposure is

$$N_{\mathrm{DSLR}} = \frac{E_{\mathrm{DSLR}}}{E_\gamma} \approx 4.35 \times 10^{14}.$$

Therefore

$$\boxed{N_{\mathrm{DSLR}} \approx 4.35 \times 10^{14}\ \mathrm{photons}}.$$

6.3 (iii) Equivalent FAST exposure

FAST receives approximately

$$3.3 \times 10^{11} \text{ photons s}^{-1}.$$

The time needed to receive the same number of photons as the DSLR exposure is

$$t_{\mathrm{FAST}} = \frac{N_{\mathrm{DSLR}}}{\dot{N}_{\mathrm{FAST}}}.$$

Thus

$$t_{\mathrm{FAST}} \approx \frac{4.35 \times 10^{14}}{3.3 \times 10^{11}} \approx 1.32 \times 10^{3} \text{ s}.$$

Therefore

$$\boxed{t_{\mathrm{FAST}} \approx 22 \text{ minutes}}.$$

The comparison is striking: although Andromeda is enormously luminous, its huge distance makes the photon flux arriving at Earth extremely small.

7 Question 7 — Solar sail

The incident light has intensity

$$I \text{ W m}^{-2}$$

and the sail has area A.

7.1 (i) Impulse delivered per second

For a photon of wavelength λ, the de Broglie relation gives

$$p = \frac{h}{\lambda}.$$

When a photon is perfectly reflected, its momentum reverses direction. Therefore the change in momentum of the photon, and hence the impulse delivered to the sail, is

$$\Delta p = 2p = \frac{2h}{\lambda}.$$

The energy of one photon is

$$E_\gamma = hf = \frac{hc}{\lambda}.$$

The number of photons arriving at the sail per second is therefore

$$\dot{N} = \frac{IA}{hc/\lambda} = \frac{IA\lambda}{hc}.$$

The impulse delivered per second is

$$F = \dot{N}\Delta p.$$

Hence

$$F = \frac{IA\lambda}{hc}\frac{2h}{\lambda} = \boxed{\frac{2IA}{c}}.$$

This is the radiation-pressure force for a perfectly reflecting sail.

7.2 (ii) Acceleration

Newton's second law gives

$$F = ma.$$

Therefore

$$ma = \frac{2IA}{c},$$

so

$$\boxed{a = \frac{2IA}{mc}}.$$

The corresponding radiation pressure is

$$P_{\text{rad}} = \frac{F}{A} = \boxed{\frac{2I}{c}}.$$

7.3 (iii) Numerical example

The spacecraft has mass

$$m = 10 \text{ tonnes} = 10^4 \text{ kg},$$

and the sail area is

$$A = 100 \text{ km}^2 = 10^8 \text{ m}^2.$$

The intensity is

$$I = 1 \text{ kW m}^{-2} = 10^3 \text{ W m}^{-2}.$$

Thus

$$a = \frac{2(10^3)(10^8)}{(10^4)(2.998 \times 10^8)} \approx 6.67 \times 10^{-2} \text{ m s}^{-2}.$$

Therefore

$$\boxed{a \approx 0.0667 \text{ m s}^{-2}}.$$

The target velocity is

$$v = \frac{c}{100}.$$

Assuming constant acceleration,

$$t = \frac{v}{a} = \frac{c/100}{a}.$$

Hence

$$t \approx \frac{2.998 \times 10^8}{100(0.0667)} \approx 4.49 \times 10^7 \text{ s}.$$

Converting to years,

$$t \approx 1.42 \text{ yr}.$$

Thus

$$\boxed{t \approx 4.49 \times 10^7 \text{ s} \approx 1.4 \text{ years}}.$$

The problem explicitly notes that this constant-acceleration treatment eventually breaks down as relativistic effects become important.

8 Question 8 — Stellar radiation and mass loss

8.1 (i) Solar luminosity and mass loss

The Earth receives a maximum solar intensity of

$$I = 1361\ \mathrm{W\,m^{-2}}.$$

At one astronomical unit,

$$r = 1.496 \times 10^{11}\ \mathrm{m}.$$

Assuming the Sun radiates isotropically, its luminosity is

$$L_\odot = 4\pi r^2 I.$$

Thus

$$L_\odot = 4\pi(1.496 \times 10^{11})^2(1361) \approx 3.83 \times 10^{26}\ \mathrm{W}.$$

Hence

$$\boxed{L_\odot \approx 3.83 \times 10^{26}\ \mathrm{W}}.$$

Over

$$t = 10^{10}\ \mathrm{yr},$$

the radiated energy is

$$E = L_\odot t.$$

Using

$$1\ \mathrm{yr} \approx 365 \times 24 \times 3600\ \mathrm{s},$$

we obtain

$$E \approx 1.21 \times 10^{44}\ \mathrm{J}.$$

Einstein's mass-energy relation gives

$$E = \Delta m c^2,$$

so

$$\Delta m = \frac{E}{c^2} \approx 1.34 \times 10^{27}\ \mathrm{kg}.$$

The Sun's mass is

$$M_\odot = 1.989 \times 10^{30}\ \mathrm{kg},$$

so the percentage mass loss is

$$\frac{\Delta m}{M_\odot} \times 100 \approx 0.067\%.$$

Therefore

$$\boxed{\text{solar mass loss over this period} \approx 0.07\%}.$$

8.2 (ii) Sirius

Sirius has

$$T_S = 9940\ \mathrm{K}, \qquad R_S = 1.711 R_\odot, \qquad M_S = 2.02 M_\odot.$$

(a) Radiation received at Earth's orbit

For the same distance from the star, the received intensity is proportional to

$$T^4R^2.$$

Therefore

$$\frac{I_S}{I_\odot} = \left(\frac{T_S}{T_\odot}\right)^4 \left(\frac{R_S}{R_\odot}\right)^2.$$

Thus

$$I_S = 1361\left(\frac{9940}{5778}\right)^4 (1.711)^2.$$

This gives

$$\boxed{I_S \approx 3.49 \times 10^4 \ \mathrm{W\,m^{-2}}}$$

or approximately

$$\boxed{34.9\ \mathrm{kW\,m^{-2}}}.$$

This is also obtained directly from the luminosity of Sirius divided by $4\pi(1\,\mathrm{AU})^2$.

(b) Luminosity of Sirius

The Stefan–Boltzmann law gives

$$L_S = 4\pi R_S^2 \sigma T_S^4.$$

Relative to the Sun,

$$\frac{L_S}{L_\odot} = \left(\frac{R_S}{R_\odot}\right)^2 \left(\frac{T_S}{T_\odot}\right)^4.$$

Therefore

$$L_S = 3.83 \times 10^{26}(1.711)^2\left(\frac{9940}{5778}\right)^4.$$

Hence

$$\boxed{L_S \approx 9.82 \times 10^{27} \ \mathrm{J\,s^{-1}}}$$

or

$$\boxed{L_S \approx 9.82 \times 10^{27} \ \mathrm{W}}.$$

(c) Lifetime

The problem asks us to assume that Sirius loses the same *fraction* of its mass over its lifetime as the Sun does during its main-sequence phase. Thus the fractional mass loss is approximately

$$f = 0.07\% = 7.0 \times 10^{-4}.$$

For Sirius, the mass converted to radiated energy is therefore

$$\Delta M_S = fM_S = f(2.02M_\odot).$$

Using $E = \Delta Mc^2$ and the Sirius luminosity from part (b),

$$t_S = \frac{\Delta M_S c^2}{L_S}.$$

Hence

$$t_S = \frac{(7.0 \times 10^{-4})(2.02)(1.989 \times 10^{30})(2.998 \times 10^8)^2}{9.82 \times 10^{27}} \approx 2.57 \times 10^{16}\,\mathrm{s}.$$

Converting to years,

$$\boxed{t_S \approx 8.2 \times 10^8\,\mathrm{yr} \approx 0.82\,\text{billion years}}.$$

This is not a realistic stellar lifetime calculation: real stellar evolution changes the luminosity, composition and mass-loss rate over time. The result follows from the specific constant-fractional-mass-loss assumption in the question.

8.3 (iii) DogStar heating elements

A $10\,\mathrm{m}^2$ panel at one astronomical unit from Sirius receives power

$$P = I_S A.$$

Using the irradiance found in part (a),

$$I_S \approx 3.49 \times 10^4\,\mathrm{W\,m^{-2}},$$

we find

$$P \approx (3.49 \times 10^4)(10) \approx 3.49 \times 10^5\,\mathrm{W}.$$

Thus the heating elements are idealised as having an available power of

$$\boxed{P \approx 3.49 \times 10^5\,\mathrm{W}}.$$

To boil one litre of water from 20°C to 100°C,

$$Q = mc\Delta T.$$

With

$$m \approx 1\ \mathrm{kg}, \qquad c = 4181\ \mathrm{J\,kg^{-1}\,K^{-1}}, \qquad \Delta T = 80\ \mathrm{K},$$

we obtain

$$Q = (1)(4181)(80) = 3.345 \times 10^5\ \mathrm{J}.$$

Using the supplied solution's power estimate,

$$P \approx 3.49 \times 10^5\ \mathrm{W},$$

the heating time is

$$t = \frac{Q}{P} \approx \frac{3.345 \times 10^5}{3.49 \times 10^5} \approx 0.96\ \mathrm{s}.$$

Therefore

$$\boxed{t \approx 1\ \mathrm{s}}.$$

This ignores heat losses, the finite heat capacity of the container, evaporation, and the energy required to maintain the water at its boiling point; it is therefore an idealised minimum heating time.

Summary of numerical results

Part	Result
1(i)	$\lambda \approx 3.26\,\mathrm{cm}$, microwave
1(ii)	$\Delta E \approx 3.81 \times 10^{-5}\,\mathrm{eV}$
2(i)	$\phi \approx 4.14\,\mathrm{eV}$
2(ii)	$I \approx 4.8\,\mathrm{A}$
3	$\lambda_{\mathrm{H_2O}} \approx 3.43 \times 10^{-11}\,\mathrm{m}$
4(i)	$V_{\mathrm{cl}} \approx 64\,\mathrm{kV}$; $V_{\mathrm{rel}} \approx 79.2\,\mathrm{kV}$
4(ii)	$\lambda_{\mathrm{cl}} \approx 4.86\,\mathrm{pm}$; $\lambda_{\mathrm{rel}} \approx 4.20\,\mathrm{pm}$
4(iii)	$d \approx 1.2 \times 10^{-10}\,\mathrm{m}$
5(i)	$\lambda_{\mathrm{H}} \approx 91.3\,\mathrm{nm}$; $\lambda_{\mathrm{He}} \approx 22.8\,\mathrm{nm}$; $\lambda_{\mathrm{Rn}} \approx 0.0124\,\mathrm{nm}$
5(ii)	Measured wavelengths: H $\approx 91.2\,\mathrm{nm}$, He $\approx 50.5\,\mathrm{nm}$, Rn $\approx 115\,\mathrm{nm}$
6	FAST $\approx 3.3 \times 10^{11}$ photons/s; equivalent DSLR exposure $\approx$ 22 min
7	$a \approx 0.0667\,\mathrm{m\,s^{-2}}$; $t \approx$ 1.4 yr to reach $c/100$
8(i)	$L_{\odot} \approx 3.83 \times 10^{26}\,\mathrm{W}$; mass loss $\approx 0.07\%$
8(ii)(a)	Sirius irradiance $\approx 3.49 \times 10^{4}\,\mathrm{W\,m^{-2}}$
8(ii)(b)	$L_S \approx 9.82 \times 10^{27}\,\mathrm{W}$
8(ii)(c)	Simplified lifetime $\approx 8.2 \times 10^{8}$ yr
8(iii)	Idealised boiling time $\approx$ 1 s